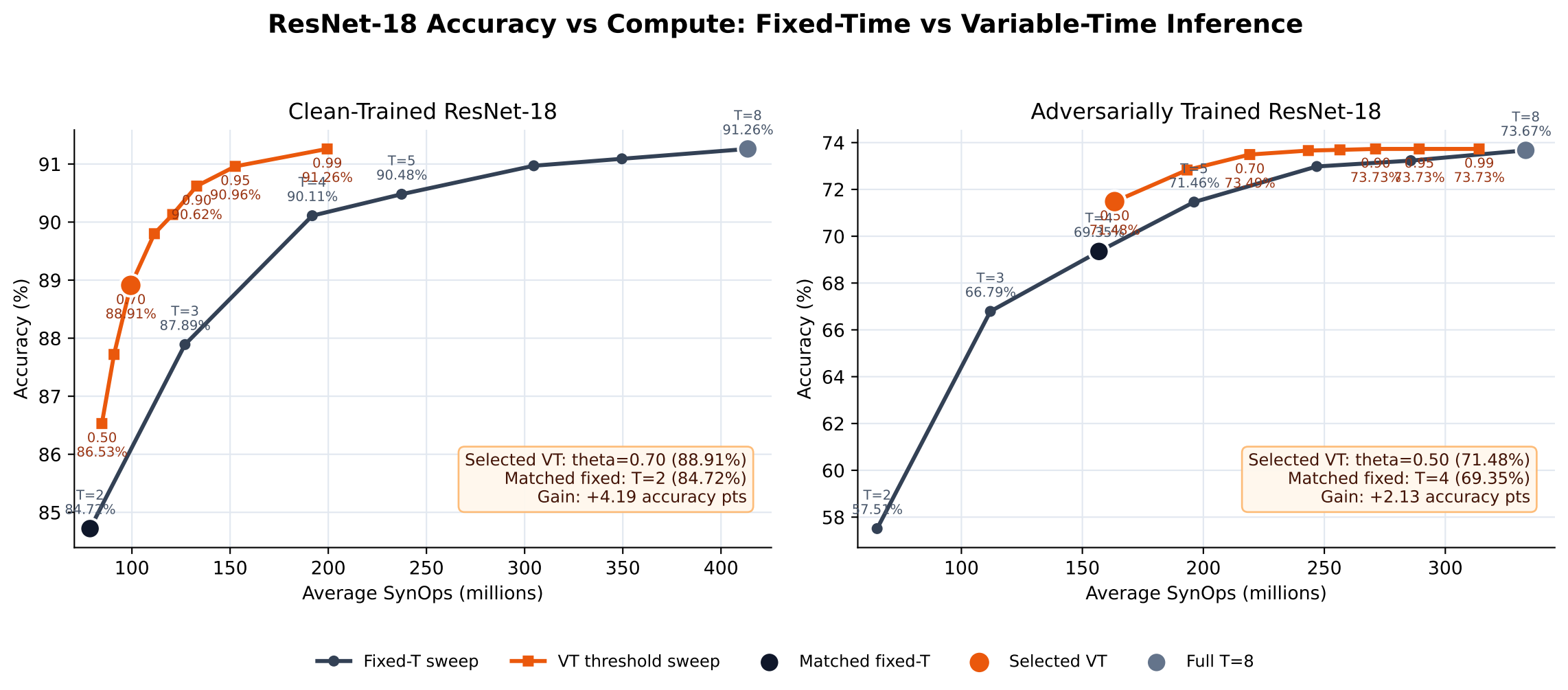
ResNet-18 Accuracy vs Compute: Fixed-Time vs Variable-Time Inference
Clean-Trained ResNet-18
Adversarially Trained ResNet-18
Accuracy (%)
Average SynOps (millions)
T=2
84.72%
T=3
87.89%
T=4
90.11%
T=5
90.48%
T=8
91.26%
0.50
86.53%
0.70
88.91%
0.90
90.62%
0.95
90.96%
0.99
91.26%
Selected VT: theta=0.70 (88.91%)
Matched fixed: T=2 (84.72%)
Gain: +4.19 accuracy pts
T=2
57.51%
T=3
66.79%
T=4
69.35%
T=5
71.46%
T=8
73.67%
0.50
71.48%
0.70
73.49%
0.90
73.73%
0.95
73.73%
0.99
73.73%
Selected VT: theta=0.50 (71.48%)
Matched fixed: T=4 (69.35%)
Gain: +2.13 accuracy pts
Fixed-T sweep
VT threshold sweep
Matched fixed-T
Selected VT
Full T=8